
SETUP GUIDE

How to Run CareOps Intelligence

A step-by-step guide to launching the full application stack — MongoDB, Neo4j, the FastAPI backend, and the Next.js frontend — locally with Docker Compose, seeding synthetic demo data, and running the test suites.

Repository: github.com/kavyasmehta/careops-intelligence

Portfolio demonstration using synthetic data. Not intended for clinical use or storage of protected health information.

1 Prerequisites

- **Docker Desktop** installed and running — this is the only hard requirement. MongoDB, Neo4j, the backend, and the frontend all run inside containers, so no local Python/Node install is strictly needed to just run the app.
- Ports `3000` , `8000` , `27018` , `7475` , and `7688` free on your machine.

These are non-default ports, chosen deliberately so this project's Mongo/Neo4j instances don't collide with any other project's databases already running on your machine.

2 Clone the repository

```
git clone https://github.com/kavyasmehta/careops-intelligence.git
cd careops-intelligence
```

3 Set up environment variables

```
cp .env.example .env
```

The defaults work out of the box — no editing required unless you want to enable the optional LLM case-summary feature (see Section 7).

4 Start the full stack

```
docker compose up --build
```

This builds and starts four containers: MongoDB, Neo4j, the FastAPI backend, and the Next.js frontend. The first run takes a few minutes while images build and dependencies install. Leave this running in its own terminal tab — Docker Compose's health checks make the backend wait for Mongo and Neo4j to report healthy before it starts.

5 Seed synthetic demo data

Open a **second terminal tab** in the same `careops-intelligence` folder:

```
./scripts/run_seed.sh
```

This populates MongoDB with ~250 synthetic clients (plus payers, providers, employees, eligibility checks, authorizations, appointments, alerts, tasks, and case notes) and builds the matching Neo4j graph from that same data.

Safe to re-run at any time. It uses a fixed random seed, so re-running always clears and rebuilds the same reproducible dataset — a good way to reset the demo to a pristine state.

6 Open the app

App (frontend)	http://localhost:3000
API docs (Swagger UI)	http://localhost:8000/docs
Backend health check	http://localhost:8000/health
Neo4j Browser	http://localhost:7475 (user <code>neo4j</code> , password from your <code>.env</code>)

On the entry screen, pick one of the three demo roles — **Operations Manager**, **Intake Specialist**, or **Authorization Specialist** — to enter the app. There is no real login: this is a portfolio project running entirely on synthetic data.

7 Optional: enable real AI case summaries

By default, the "Generate case summary" feature on a client's profile uses a deterministic template — no API key needed, and it always works. To use a real LLM instead:

1. Edit `.env`: set `ENABLE_LLM_SUMMARY=true` and `LLM_API_KEY=<your key>`.
2. Install the extra dependency: `docker compose exec backend pip install anthropic` (or add `anthropic` to `backend/requirements.txt` and rebuild).
3. Restart the backend: `docker compose restart backend`.

The LLM path never runs unless both the flag and the key are set. With no key, the app always falls back to the template summary — it never silently fails.

8 Running the test suites

Backend — 48 Pytest tests

```
cd backend
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
pytest
```

Frontend unit tests — 14 Vitest tests

```
cd frontend
npm install
npm run test
```

End-to-end test — Playwright

Requires the full Docker Compose stack to be up and seeded (Sections 4–5):

```
cd frontend
npm run test:e2e
```

This test resolves a real alert as part of its workflow. Re-run `./scripts/run_seed.sh` afterward if you want the demo data pristine again.

9 Stopping the app

```
# In the terminal running Docker Compose:
Ctrl+C
docker compose down
```

Add `-v` (`docker compose down -v`) to also wipe the Mongo/Neo4j data volumes if you want a completely fresh start next time.

10 Troubleshooting

Symptom	Fix
Port already in use	Something else is using 3000 / 8000 / 27018 / 7475 / 7688. Stop that process, or remap the port in <code>docker-compose.yml</code> .
Backend container won't start	Run <code>docker compose logs backend</code> . Usually Mongo/Neo4j hadn't finished starting; a <code>docker compose restart backend</code> once the others are healthy resolves the race.
Pages load but are empty	You haven't run Section 5 (<code>./scripts/run_seed.sh</code>) yet.

CareOps Intelligence — portfolio demonstration using synthetic data. All screenshots, data, and organizations referenced are fictional.